6	**(a)**	extension is proportional to force / load	B1	[1]
	(b)	$F = mg$	C1	
		$x = (mg / k) = 0.41 \times 9.81 / 25 = (4.02 / 25)$	M1	
		$x = 0.16\,\text{m}$	A0	[2]

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(c) **(i)** weight and (reaction) force from spring (which is equal to tension in spring) B1 [1]

(ii) F – weight or $0.06 \times 25 = ma$ C1
$F = 0.2209 \times 25 = 5.52$ (N) or $0.22 \times 25 = 5.5$
$a = (5.52 - 0.41 \times 9.81) / 0.41$ or $1.5 / 0.41$ and $(5.5 - 4.02)$ C1
$a = 3.7\ (3.66)\ \mathrm{m\,s^{-2}}$ gives $3.6\ \mathrm{m\,s^{-2}}$ A1 [3]

(d) elastic potential energy / strain energy to kinetic energy and gravitational potential energy B1
stretching / extension reduces and velocity increases / height increases B1 [2]

7 **(a)** ${}^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{1}_{1}\mathrm{p} + Q$